

Paige Randall North
Assistant Professor
Department of Mathematics
Department of Information and Computing Sciences
Utrecht University
Buys Ballot Gebouw
Princetonplein 5
3584 CC Utrecht
Netherlands

To whom it may concern:

This letter expresses my support for the research proposal submitted by Thorsten Altenkirch, Ulrich Buchholtz, and Nicolai Kraus entitled “Foundations of Directed Type Theory”.

Formulating an expressive and useable directed type theory is an important problem in current type theoretic research. Indeed, many proposals have appeared in the literature in the last 15 years from myself, the authors of this proposal, and others.

Directed type theory is important because of how much science relies on the mathematics that we seek to formalize with directed type theory. Much of that is category theory and higher category theory: indeed the foundations of modern algebraic mathematics relies on category theory, and therefore the methods of many other sciences such as physics and theoretical computer science, especially programming language theory. But that also includes directed homotopy theory, with other important applications in computer science, biology, and physics. With a computer proof assistant based on directed type theory, we could formalize and verify all of this mathematics. Additionally, such a computer proof assistant would ease the load of the complexity inherent in these applied and higher mathematical techniques, allowing us to apply them at scale.

From a more specific, programming languages perspective, a directed type theory would allow us to internalize much metatheory and allow us to derive “theorems for free”.

As a collaborator of this project, I will assist with WP1 and WP2, helping to designing the syntax and semantics of 1-dimensional and ∞ -dimensional directed type theory. This aligns with my prior expertise and papers in directed type theory.

I plan to spend about 10 days a year collaborating on this project, which corresponds to an in-kind contribution of £5000.

Yours sincerely,

Paige Randall North